Notes on Distributions, Sobolev Spaces and Weak Solutions

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The main objective of this text is to compile a everything I've read or discussed about weak solutions on PDEs and the prerequisites for understanding this topic.

Summary

CHAPTER 1

Motivation

We will begin by studying the solution of the following equation:

$$u_t + uu_x = 0, \text{ for } x \in \mathbb{R} \text{ and } t > 0$$

This is comunly known as the inviscid Burger's equation, a simplified model for fluid mechanics. So, first, we are going to try to find regular solutions for this equation by the method of characteristic. Considering $u(x,t)$ to be a smooth solution for our equation satisfying the initial condition problem

$$u(x,0) = u_0(x)$$

Where $u_0(x)$ is a function of x. By defining the chatacteristic curves:

$$\begin{cases} \dot{x}(t) = u(x(t), x),\ x(0) = x_0 \\ \dot{y}(t) = 1,\ y(0) = y_0 \end{cases} \tag{1}$$

It is given that $y = t$. By calculating the derivative of $u(x(t), t)$ we obtain:

$$\frac{\mathrm{d}u}{\mathrm{d}t} = \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x}\frac{\partial x}{\partial t} = 0$$

From this, we can get two conclusions: u is constant along the curve $(x(t), t)$, which implies that $\dot{x}(t) = u_0(x_0)$. By solving this last ODE, we obtain that:

$$x = x_0 + u_0(x_0)t$$

And for all points in this curve $u(x,t) = u_0(x_0)$. By convenience, we will choose this as our initial curve u_0:

Using the initial condition given by u_0, the characteristic curves are:

We can clearly see that all of the curves are intersecting one another, so the value of $u(x,t)$ is not even well defined at the intersection points